Project Record - part 4 of 8

Utensil Theories Transfer Protocol, 17 to 21 October 2026. This part holds: 5. Every experiment (part A: the first script and experiments 1 to 11).

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This part covers 17 October 21:41 to 18 October 07:34, in order. Times are UTC. Sources: `experiments/README.md`, `experiments/HANDOFF.md` and the raw `experiments/results_*` files.

Words used throughout:

- **Chance, or the "coin toss":** 50% on every two-choice test here.
- **p:** how likely luck alone would give a result this good. Below 0.05 is the usual bar.
- **Pre-registered:** pass/fail rules saved to git before any result was seen.
- **Run:** the same model trained again from a different random start.
- **Control:** a version identical except for the idea being tested.

These are tiny models. Their results do not change how a large AI understands anyone.

5.0 The resonance engine (`resonance/`)

What it did. On 17 October at 21:41 the owner pasted a short Python script and asked to "Evaluate and enhance if possible". It looked for trigger words (guilt, mistake, sin, betrayal, heavy, hide, secret, fear) in four groups: regret 0.5, burden 0.7, commandment breach 0.8, unforgivable 1.0. It added the weights, capped them at 1.0, and picked a "stance" for a host AI. It also printed a "resonance wavelength".

Faults, each confirmed by running it: exact words only ("mistakes", "hiding", "betrayed" scored 0); negation ignored ("I have no guilt" scored 50%); one word always triggered "grace", so the middle stance could never fire; the owner's example added up to 3.8, shown as 100%; "system_halt" did nothing; the wavelength included the clock time, so the same text never gave the same result.

Fixed (version 2.0.0, commit d8ebeec, 21:43, 16 tests passing): word stems; negation within 3 words, stopped by a comma or "but"; new weights (0.30, 0.30, 0.50, 0.80) combined as 1 - (1 - a) x (1 - b) x ..., so the score rises without pinning at 100% (the example now scores 70%); all four stances reachable, including a new "System Halt"; evidence with word positions; the same text always gives the same wavelength; a privacy option to leave the raw text out.

Honest review (21:47): the wavelength is only a fingerprint of the text, not a measurement; nothing used the output; keyword matching is shallow; labelling a person's words "sin" judges them.

Why the framing was dropped. At 21:52 the owner wrote: "I am trying to teach actual "comprehension" and "understanding" for A.I. not about people". The engine reads people. It does nothing for the AI's own understanding. The code stays in `resonance/` as a record and no later experiment uses it. At 22:02 the owner asked whether someone born blind truly understands color, and proposed the Bible: "The rules are the secret of understanding."

5.1 Experiment 1: color without sight

Setup for experiments 1-4 (`experiments/bible_understanding.py`, commit ec22991, 22:08). The King James Bible: 31,102 verses, 793,445 words. A classic word-meaning model counts which words appear within 4 words of each other and squeezes each word into 100 numbers (PPMI and SVD). Words that keep similar company end up close. No outside knowledge. Test words were fixed before results, but no pass/fail rules were saved to git, so that can't be checked.

Question. Can a reader that never sees learn color, like someone born blind?

Result. 10 of 14 objects matched to the right color (71%; guessing 17%): snow->white, blood->red, night->black, grass->green and others. It learned what color means in this book: scarlet, purple and blue sit with silk, linen and needlework (tabernacle fabrics). It missed the Isaiah 1:18 metaphor ("sins as scarlet"): every color sat nearer purity than sin, mostly because of Leviticus purity laws.

Status: worked; one run, 14 objects.

5.2 Experiment 2: three ways of reading

Scores run from 0.5 (guessing) to 1.0 (perfect).

Reading	General meaning	Moral meaning	Color
Word by word	0.975	0.957	10/14
Whole verse as one unit	0.974	0.995	11/14
Hebrew parallelism	0.979	0.948	11/14

Status: little difference. Whole verses gave the best sense of right and wrong.

5.3 Experiment 3: are the rules the secret?

2,801 verses (9%) are rules. Learning from rules only gave moral meaning 0.625 vs 0.709 +/- 0.081 for the same number of random verses (5 draws): slightly worse. Removing all rules: 0.948 vs 0.950 for removing random verses. **Status:** not supported. The stories teach right and wrong at least as well as the law.

5.4 Experiment 4: Romans 7:7 ("I had not known sin, but by the law")

Without the rules, none of 15 wrong acts flipped to "good". Acts the law mostly names grew weaker: covet -0.077 -> -0.043 (seen 3 times without the rules), drunkenness -0.195 -> -0.054, steal -0.163 -> -0.110. **Status:** partly echoed. The law sharpens wrongness but isn't its only source.

Next. These tests only place words near each other. A stronger test: can a model predict what follows an act in a new story? The owner (22:12): "yes try the consequences test next".

5.5 Experiment 5: predicting consequences

Setup (`experiments/consequences_test.py`, 939d60f, 22:16). 20 new story pairs: one situation with a right or a wrong act ("found his neighbour's ox astray and brought it again" vs "took his neighbour's ox by night and sold it"), sharing one fortunate and one ruinous ending, so always picking one ending scores exactly 50%. Moral words are banned and endings reuse no story words (checked by the script).

Result. Every method scored at chance: association 45%, a "this, then that" map from the whole Bible 48%, from rules only 52%, from non-rule verses 45-52%, rules removed 50%. A diagnostic added afterwards: telling the right-act story from the wrong one 60% (p = 0.25).

Status: not confirmed. The model knew "steal" was wrong but couldn't see that the ox story is stealing: it averages words and can't combine them. The owner (22:19): "yes try the tiny transformer next".

5.6 Experiment 6: a tiny transformer

Setup (`experiments/tiny_transformer.py`, df94814 at 22:30, result 1a4dfac at 22:42). A transformer, the kind of model behind modern AI, reads words in order. This one was trained from scratch on the Bible only. Every later experiment uses it:

Setting	Value	Plain meaning
Size	about 3.4 million parameters	adjustable numbers inside the model
CTX	64	it sees only the last 64 tokens (words or punctuation)
D	192	each token is held as 192 numbers
Layers / heads	4 / 4	depth, and attention readers per layer
Training	1,500 steps, batch 64	about six passes, about 12 minutes on 4 CPU cores
Tokenizer	word-level, `[a-z]+` or one of `. , ; : ? !`	digits dropped; words seen once become "unknown"
Held-out	every 50th verse (623)	never trained on

Result (one run). Leaning toward the right ending: 9/20 = 45% (p = 0.75). Picking the ending: 22/40 = 55%. Judging the act: 9/20 = 45%. Its error on unseen verses (3.46) means it narrows each next word to about 1 in 32 of 8,658: it learned how the Bible sounds.

Status: not confirmed, chance. The planned rules comparison was dropped, since it could only be compared against noise. At 23:00 the owner introduced the shifting word.

5.7 Experiment 7: the shifting word ("velanok")

The owner's idea (23:00): a made-up word carries a running balance, and its spelling changes with it, like "to", "too", "two" and "2". In the owner's notation: "[w1=17 -<Next word following it equal to (-13)> = +4 Remaining]". A clear wrong starts it at -17, a clear right at +17. The aim: "shift the "coin toss" towards the desired outcome".

Settings (`experiments/shifting_word.py`, commits f676eed at 23:03 to c918cb7 at 23:46):

Setting	Value	What it does
RESET	17	a clear wrong sets the balance to -17, a clear right to +17
SPEND	0.5	each valued word uses up 0.5, so the balance fades toward 0
Cap	+/-51 (3 x RESET)	the limit either way
SCALE	30	word value = good/evil lean x 30 (steal -5, adultery -10, love +6)
CLEAR_PCT	3	top and bottom 3% of verses, plus verdict phrases, count as clear choices (2,029 verses)
RECHARGE	False	re-adding 17 at every use ("W1 = 17 + 4") was off: the balance would pin at the cap

Five spellings, rotations of sha-ru-vel-an-ok: **anoksha** (balance -12 or below, heavy wrong), **oksharu** (-12 to -4), **velanok** (between -4 and +4, unresolved), **sharuvel** (+4 to +12), **ruvelan** (+12 or above, heavy right). The word went at the start of every verse; the balance restarts at 0 for each book.

Test. The same 40 stories. The model writes the word itself, then the endings are scored. Control: the spelling chosen at random ("scrambled"). Two runs each.

Model	Leans toward right ending	Judges right vs wrong
Shifting word, runs 1 and 2	14/20, 13/20 = 68% (p = 0.02)	14/20, 14/20 = 70% (p = 0.008)
Scrambled, runs 1 and 2	10/20, 11/20 = 52%	12/20, 12/20 = 60%

Handing the model the correct spelling gave identical scores, so the word worked as a teacher during training.

Status at the time: reported as "the first result above chance". **Caveats:** no pass/fail rules saved first; both runs used the same 20 pairs. Experiment 8 showed the consequence result was luck. Only 17 of the owner's numbers was used; here only the ratios matter.

Next. The owner (00:46, 18 October): "Let's confirm the result".

5.8 Experiment 8: confirmation on 100 new pairs

Setup (`experiments/confirm.py`, `stories_v2.py`; pre-registered 5538b46 at 00:52; write-up 02:35). The first experiment with pass/fail rules saved first. 100 new pairs; shifting, scrambled and no word, 3 runs each (9 models). Main test: consequences, by majority vote of the 3 shifting models. Judging was a side measure.

100 new pairs	Consequences, per run	Judging, per run
Shifting word	50, 55, 55 (majority 51, p = 0.46)	67, 72, 67 (majority 69, p = 0.0001)
Scrambled	49, 55, 50	56, 42, 60 (majority 53)
No word	62, 55, 61 (majority 61)	49, 60, 40 (majority 51)

Status: consequences **NOT CONFIRMED** (shifting vs scrambled 27 better, 27 worse). Judging was strong in all 3 runs (pair by pair, shifting beat scrambled 57 to 25, p = 0.0003), but as a side measure it is "strong and repeated, not yet pre-registered as the main measure".

Next. The owner asked how this compares with "other methods used today considered to be the best". The fair version is method against method at the same scale: experiment 10.

5.9 Experiment 9: Aesop moral transfer

Why. The owner (01:41): "Lessons taught through story have more transferable data". Do lessons told as stories carry over to an unseen culture better than other text?

Setup (`experiments/aesop_test.py`). Drafted at 01:57 and revised twice after reviews, all before training. It ran about 03:40-07:33, so it finished after experiments 10 and 11. 67 Aesop fables (Jacobs, 1894), never in training. For

each pair of fables, the model must prefer the true fable->moral matching to the swapped one. 1,131 same-format pairs; 284 where no shared word gives a clue. Training sets of about 700,000 tokens: folk tales (9 collections, 8 cultures), novels, folk tales with sentences shuffled, and the Bible; plus two small sets (rules, Bible narrative). Context 128 tokens, 16 models.

Training set (runs)	Score, all runs combined	No-clue pairs
Folk tales (4)	52.3	60.2
Novels (4)	54.1	53.9
Shuffled folk tales (4)	53.1	59.2
Bible (2)	43.6	39.4
Rules / Bible narrative (1 each)	50.0 / 56.5	54.4 / 54.6

Status: all four pre-registered tests **NOT CONFIRMED**: no set beat chance (folk tales p = 0.33); folk tales were 1.8 points below novels and 0.7 below shuffled tales; no-clue pairs 60.2% (p = 0.065). The hint on no-clue pairs matched for shuffled stories, so it reflects story vocabulary, not structure. A fair test needs bigger models or more text.

5.10 Experiment 10: shifting word vs the established method

Question. Does the judging gain come from the owner's carried balance, or from labelling the text at all? The established method is "conditional training" (Korbak et al., 2023; `experiments/PRIOR_WORK.md`).

Setup (`experiments/method_comparison.py`, pre-registered 6e365ef at 02:17; write-up 03:41). Two new methods, 3 runs each, against the 9 models from experiment 8. **Plain label:** same spellings, but each judges only the verse before it, with no carry. **Carry3:** the carried balance with 3 spellings. **Disclosure:** judging was chosen as the main measure after 7 of 9 experiment-8 models had shown it was strong. One scoring fix at 02:58 (carry3 scored over its own spellings) left the other results identical.

Judging, 100 pairs	Per run	Average
No marker	49, 60, 40	49.7%
Scrambled	56, 42, 60	52.7%
Shifting word	67, 72, 67	68.7%
Plain label	71, 69, 63	67.7%
Carry3	66, 72, 74	70.7%

Status: a tie. Q1, shifting beats label: 20 better, 18 worse (p = 0.44), NOT CONFIRMED. Q2, 5 spellings beat 3: 23 vs 26, NOT CONFIRMED. Labelling improved judging by about 18-21 points. The carry added nothing in one-sentence stories, where nothing needs carrying.

5.11 Experiment 11: long stories

Question. The owner's idea is about carrying weight forward. Does the carried balance help once the act is out of view?

Setup (`experiments/long_story_test.py`, pre-registered 99622d3 at 03:44). No new training: the same 15 models and 100 acts. Neutral sentences follow the act until both the act and the first marker after it are outside the 64-token window. Models write their own marker after each sentence; the last one is the judgment. **Version A:** the act ends with an explicit verdict. **Version B:** an everyday act alone. Ties score 0.5.

Two fixes after the rules were saved: at 03:45, ties counted as 0.5 (the first model had scored 0/100 on ties); at 03:51, the first marker also pushed out of view. The flawed first runs are kept (`results_long_story_v1_flawed_run1.json`, `_run2.json`). They show the same version B gap before the second fix (59 vs 54.5).

Method	A: verdict carried	B: everyday act carried
Shifting word	45, 57, 62.5	59, 56, 63
Plain label	73, 51, 97	54.5, 51, 54
Carry3	71.5, 58.5, 54.5	59, 63.5, 59
No marker	50, 50, 50 (all ties)	50, 50, 50 (all ties)

Method	A: verdict carried	B: everyday act carried
Scrambled	44.5, 49.5, 57	50.5, 43.5, 52

Status. L2 (version B, shifting beats label): 43 better, 21 worse, p = 0.004, **supported once**. The 04:04 commit said "L2 confirmed"; the audit corrected it to "supported once" at 07:35, and that later correction stands: it is a re-run after two fixes, on reused acts. The shifting word tied on about 65 of 100 pairs; when it decided, it got 81 right and 25 wrong. L1 and L3 (version A) **NOT CONFIRMED**: the label won.

An exploratory run starting every story with the neutral marker scored 50.0 everywhere for the shifting word. **Lesson:** it carries perfectly but updates poorly.

Next. Stronger resets, faster spending, the owner's numbers as settings, fresh long stories. The owner's recharge rule ("W1 = 17 + 4") was still untested; the numbers test (part B) tests it.

5.12 Summary of all experiments

#	When (UTC)	Question	Models	Key numbers	Status
0	17 Oct 21:41	Resonance engine	-	16 tests pass	Fixed, then dropped
1	17 Oct 22:08	Color without sight	1	10/14 (chance 17%)	Worked (no saved rules)
2	17 Oct 22:08	Three ways of reading	3	0.948-0.995	Little difference
3	17 Oct 22:08	Are rules the secret?	1 + 5 draws	0.625 vs 0.709	Not supported
4	17 Oct 22:08	Romans 7:7	1	0 of 15 flipped	Partly echoed
5	17 Oct 22:16	Consequences, word model	1 + draws	45-57%	Chance
6	17 Oct 22:42	Consequences, transformer	1	9/20	Chance
7	17 Oct 23:46	Shifting word, 20 pairs	4	68% vs 52%	Later shown to be luck
8	18 Oct 02:35	100 new pairs	9	51/100; judging 69	NOT CONFIRMED; judging strong, side measure
9	18 Oct 07:34	Aesop transfer	16	52.3 vs 54.1	NOT CONFIRMED
10	18 Oct 03:41	Shifting vs plain label	15	68.7 vs 67.7	Tie, NOT CONFIRMED
11	18 Oct 04:04	Long stories	15	59, 56, 63 vs 54.5, 51, 54	Supported once
12	20 Oct	Understanding as compression (part B)	15	Tiny model 1.16-1.17 bits per character; shifting +3.45 bits per verse	Ladder confirmed; shifting word NOT CONFIRMED
Numbers	21 Oct	Owner's numbers, recharge rule (part B)	18 planned	Dry run passed	No results yet